Design and Development of a Rule-Based AI-Powered Student Productivity and Academic Scheduling System

APT 3065A

Spring Semester 2026

Prof. Fredrick Michael Ogore

Victor Chege - 662833

CHAPTER ONE

INTRODUCTION

In recent years, the higher education learning environment has changed due to the rapid advancement of digital technologies. Universities increasingly rely on digital platforms to support learning, communication, assessments and academic management. However, despite technological integration, students continue to face challenges in effective management of academic responsibilities such as assignments, projects and assessments. Effective time management and structured planning remain critical determinants of academic success(Britton & Tesser, 1991).

Artificial intelligence has been a transformative technology capable of enhancing decision making, automation and personalization across different industries including education. In educational contexts, AI driven systems have been applied to adaptive learning, intelligent tutoring systems and learning analytics to improve student outcomes (Holmes, Bialik & Fedal, 2019). Despite this progress, insufficient research exists on AI enabled systems designed specifically for prioritizing student tasks and generating customized academic schedules.

This project proposes the design and development of an AI-powered student productivity system that integrates rule-based intelligent logic to assist university students in prioritizing academic tasks and generating adaptive study schedules.

BACKGROUND OF THE STUDY

Students' academic success depends significantly on their capacity to manage their own learning. According to self regulated learning theory, high performing students consistently plan their work, track their progress, and assess their outcomes (Zimmerman, 2002). Effective time management stands out as an essential component of this process. Studies confirm that students who implement organized time management approaches earn better grades than those lacking such systematic methods (Britton & Tesser, 1991).

With the widespread adoption of digital technologies, students increasingly rely on productivity applications such as Todoist, Notion, and Microsoft To Do to organize academic responsibilities

through task lists, deadline reminders, and calendar integration. These platforms provide structured organizational features that assist users in tracking assignments and upcoming obligations. However, such systems largely depend on manual prioritization, requiring students to determine task urgency and importance independently. Research on self-regulated learning indicates that effective academic performance depends on structured planning and strategic time allocation, skills that many students struggle to consistently apply without guided support (Zimmerman, 2002).

More recently, AI enhanced scheduling tools such as Motion and Reclaim.ai have introduced automated calendar management and task rescheduling features. While these systems demonstrate the growing integration of AI into productivity software, they are primarily designed for professional and corporate environments rather than academic contexts. They do not incorporate features tailored to university coursework, such as course-based workload balancing or examination-period adjustments. Similarly, Learning Management Systems (LMS) such as Moodle and Blackboard are widely used in higher education institutions; however, their primary function is content delivery, communication, and assessment management rather than personalized study scheduling (Siemens, 2013).

In the African and Kenyan education context, educational technology platforms such as ShulePro and Cloud School System focus mainly on institutional administration, including timetabling, attendance monitoring, and financial management. These systems support school operations but do not provide individual students with intelligent productivity tools or AI driven task prioritization mechanisms. This landscape demonstrates that while digital and AI enabled platforms exist, there remains a gap in systems specifically designed to support structured, student-centered academic scheduling through transparent and adaptive prioritization logic.

The growing integration of Artificial Intelligence in education has largely concentrated on adaptive learning environments, intelligent tutoring systems, and performance analytics (Holmes, Bialik, & Fadel, 2019). While these applications demonstrate the potential of AI to enhance educational outcomes, limited attention has been given to AI driven workload scheduling and structured academic time management. This gap presents an opportunity to design a system that integrates transparent, rule-based intelligent logic into a web based platform specifically tailored to support university students in prioritizing academic tasks and generating adaptive study schedules.

PROBLEM STATEMENT

University students are required to manage multiple academic responsibilities simultaneously, including assignments, group projects, presentations, quizzes, and examinations. These tasks

frequently overlap within limited time frames, creating significant academic pressure. Despite the availability of digital tools, many students struggle with effective time allocation and structured prioritization of academic activities.

Empirical research demonstrates that poor time management is directly associated with lower academic performance and increased stress levels (Britton & Tesser, 1991). Furthermore, self-regulated learning theory suggests that students who lack structured planning strategies experience difficulty achieving academic goals (Zimmerman, 2002). While digital productivity tools provide reminders and task tracking functionalities, they do not function as intelligent decision-support systems capable of automatically evaluating urgency, workload distribution, and task complexity.

From an Information Systems perspective, decision support systems (DSS) are designed to assist users in making structured and semi-structured decisions by analyzing relevant data and applying logical models (Power, 2002). Recent developments in learning analytics further demonstrate the importance of data-driven systems in supporting academic decision-making processes within higher education (Viberg, Hatakka, Bälter, & Mavroudi, 2020). However, most existing academic productivity tools do not integrate such analytical or decision-support capabilities into personalized workload scheduling. Instead, students remain responsible for manually determining task priority, which increases cognitive load and susceptibility to procrastination.

Similarly, Learning Management Systems focus primarily on instructional delivery and assessment management rather than personalized academic scheduling (Siemens, 2013). The absence of an integrated AI-driven academic decision-support system that automates task prioritization represents a significant technological and practical gap in higher education environments.

As a result, students frequently:

- Having trouble determining which tasks require immediate attention
- Misjudge the time required to complete complex assignments
- Experience last-minute academic pressure due to unclear prioritization
- Struggle to balance competing deadlines during peak academic periods

The absence of an integrated, AI driven academic decision-support system that automates task prioritization and generates adaptive study schedules represents a significant technological and practical gap in higher education. There is therefore a need to design and develop a web-based system that integrates rule-based Artificial Intelligence logic to provide transparent, consistent, and personalized academic scheduling support for university students.

MAIN OBJECTIVE

To design and develop an AI powered web based student productivity system that integrates intelligent task prioritization and adaptive scheduling to improve academic time management among university students.

SPECIFIC OBJECTIVES

1. To design a secure and user-friendly web-based platform for managing academic tasks and deadlines.
2. To develop a rule-based AI prioritization algorithm that evaluates tasks based on urgency and complexity.
3. To integrate automated schedule generation that dynamically adjusts study plans when tasks are updated or added.
4. To enhance student productivity and academic organization through intelligent decision-support mechanisms.

SIGNIFICANCE OF THE STUDY

The significance of this project is academic, practical, and professional in nature.

First, the system addresses a common and critical challenge faced by university students, including myself which is ineffective time management and difficulty in prioritizing overlapping academic tasks. Throughout my academic experience, I have observed the challenges students encounter when balancing assignments, examinations, and project deadlines within limited time frames. By integrating intelligent prioritization logic, the proposed system reduces reliance on manual planning and supports structured academic decision making. It therefore responds not only to a theoretical problem identified in literature but also to a real and experienced academic need.

Second, the project contributes to the field of Information Systems by demonstrating the practical application of Design Science Research principles in developing a purposeful digital artifact (Hevner et al., 2004). The system represents an implementation of rule based Artificial Intelligence within a web-based architecture, showing how analytical logic can be embedded into software solutions to solve real world educational problems.

Third, within the context of higher education institutions, the system provides a scalable prototype that could be extended into institutional academic support services. As universities increasingly explore AI enabled tools to enhance student success, this project aligns with global trends in educational technology while addressing a localized gap in structured academic productivity systems tailored for students.

Finally, the development of this project enhances my technical and professional competencies in full-stack web development, database design, system architecture, AI logic implementation and integration. By designing and building the system independently, I strengthen my practical skills within the Information Systems discipline while producing a solution that holds both academic and societal relevance.

CHAPTER TWO

LITERATURE REVIEW

This chapter reviews existing scholarly work and operational systems related to student productivity, intelligent scheduling, and Artificial Intelligence(AI) in education. It presents what has been developed globally, in Africa, and in Kenya, then identifies limitations of existing approaches and clarifies the research gap that informs the proposed AI-powered student productivity and study planning system.

Global Perspective: Student Productivity and Intelligent Scheduling Systems

Globally, research has demonstrated that students' academic success is strongly influenced by effective planning, time allocation, and self-regulated learning practices. Self-regulated learning literature emphasizes planning and time management as core mechanisms through which learners control performance outcomes (Zimmerman, 2002). However, many students struggle to consistently apply these skills without structured support, which has motivated the development of digital productivity systems and intelligent scheduling approaches.

Several research prototypes and algorithmic scheduling systems have been proposed to support structured planning. For example, early academic work such as EduPlan explored intelligent time management support for students by providing structured planning assistance (Moka, 2010). More recent research has advanced algorithmic scheduling by integrating rule-based methods and optimization approaches. A notable example is the Automated Study Plan Generator that combined rule-based logic with optimization (knapsack-based selection) to generate structured study plans under constraints, demonstrating that rule-based systems can effectively support academic planning decisions without heavy computational requirements (Alotaibi et al., 2022).

In parallel, AI in education research has expanded significantly in recent years, focusing largely on intelligent tutoring, adaptive learning, and learning analytics. Systematic reviews show AI can personalize learning and support decision-making; however, most studies emphasize learning enhancement and monitoring rather than automated workload scheduling for daily student productivity (Garzón et al., 2025).

Existing global platforms: Students commonly use tools such as Todoist, Notion, and Microsoft To Do for task listing and reminders, while some AI-enhanced planners such as Motion and Reclaim.ai offer automated calendar scheduling. However, these platforms typically emphasize general productivity and professional workflows rather than student-specific academic scheduling (e.g.,

course workload balancing, exam-period planning). This gap is reflected in the research direction, where intelligent scheduling solutions for students are still less developed compared to tutoring or analytics systems.

African Perspective: AI Adoption in Higher Education and Student Support

Across Africa, AI adoption in higher education is emerging but uneven. A systematic review of AI adoption in African higher education in the last four years reports that AI is most applied to learning support, administrative efficiency, and skills development, but also highlights persistent barriers including infrastructure, limited institutional capacity, and ethical concerns (Maluleke, 2025).

Within African contexts, AI implementations in universities largely focus on institutional priorities such as improving teaching support, assessment, and administrative management rather than developing student centered scheduling systems. A South African focused systematic review similarly concludes that while AI is being explored across higher education, adoption remains constrained by policy, training, and governance factors (Khoalenyane, 2024).

This indicates that although AI is increasingly recognized in African higher education, practical implementations that directly support individual student productivity through intelligent scheduling and prioritization remain limited.

Kenyan Perspective: Digital Learning Systems and AI Use

In Kenya, universities have widely adopted Learning Management Systems (LMS) to support blended and online learning. However, evidence suggests that LMS adoption primarily supports content delivery, communication, and assessment administration rather than personalized scheduling or workload planning. For instance, Kenyan university students perceive LMS features as useful for supporting self regulated learning, but LMS platforms do not typically provide intelligent scheduling functions or automated prioritization for individual academic workloads (Araka et al., 2021).

Additionally, empirical work on AI tools in Kenyan higher education indicates that AI is being explored in areas such as intelligent tutoring, adaptive learning, and learning analytics; nonetheless, accessibility and systematic integration remain challenges (Matere, 2024).

At the institutional technology level, Kenya also has education platforms focused largely on administration (e.g., timetabling, attendance, fee management). These systems support institutional operations but do not provide individualized student decision-support mechanisms for planning study time, prioritizing tasks, or adapting schedules during peak workload periods.

EXISTING SYSTEMS AND THEIR LIMITATIONS

A variety of digital systems have been developed to support academic organization and productivity; however, their effectiveness in providing intelligent academic scheduling remains limited. Task management and digital planner applications such as Todoist, Notion, Microsoft To Do, and Google Calendar are widely used by students to organize assignments and set reminders. These platforms provide structured interfaces for task listing, deadline tracking, and calendar visualization. While they enhance organization and reduce the likelihood of missed deadlines, they largely depend on manual prioritization. Students must independently determine which tasks require immediate attention, how much time should be allocated to each activity, and how to balance competing academic demands. As such, these tools function primarily as organizational aids rather than intelligent decision-support systems capable of dynamically analyzing workload intensity or task complexity.

Learning Management Systems (LMS) such as Moodle and Blackboard are extensively implemented within higher education institutions to facilitate content delivery, communication, and assessment management. Although LMS platforms centralize academic information and provide access to assignment deadlines, their core purpose is instructional administration rather than personalized academic planning. Research indicates that while LMS features may indirectly support self-regulated learning, they do not incorporate automated scheduling or adaptive prioritization mechanisms tailored to individual student needs (Araka et al., 2021). Consequently, students remain responsible for structuring their own study plans without system-generated guidance.

In parallel, AI applications in education have expanded significantly in recent years. Intelligent tutoring systems, adaptive learning platforms, and learning analytics dashboards have demonstrated the potential of AI to personalize instruction and predict academic performance trends (Holmes et al., 2019; Viberg et al., 2020). However, the primary focus of these systems remains instructional enhancement and performance monitoring rather than operational academic workload management. They do not typically address the day-to-day challenge of allocating study time across multiple courses and deadlines.

Within the African and Kenyan context, digital transformation in higher education has largely emphasized institutional efficiency. Platforms such as ShulePro and Cloud School System focus on administrative processes including timetabling, attendance tracking, and financial management. While these systems improve institutional operations, they do not provide individualized productivity tools that assist students in intelligently structuring their academic tasks. As a result, despite the availability of digital and AI enabled platforms, students continue to lack structured, adaptive decision-support mechanisms specifically designed for academic scheduling.

The review of existing systems therefore reveals a consistent pattern: current tools either support basic task tracking, instructional delivery, or performance analytics, but none adequately integrate intelligent prioritization with personalized study schedule generation. This limitation highlights a clear and persistent gap in the design of student centered academic productivity systems.

RESEARCH GAP AND UNIQUE CONTRIBUTION OF THE PROPOSED SYSTEM

The literature and system analysis indicate that while digital productivity tools are widely available and AI adoption in education is growing, there remains limited implementation of systems that directly support structured academic decision making through intelligent scheduling. Existing productivity applications rely on manual prioritization, Learning Management Systems focus on content administration, and AI based educational technologies emphasize tutoring and analytics rather than workload planning.

Furthermore, within African and Kenyan higher education environments, the integration of AI into student centered productivity systems remains underdeveloped. Although research recognizes the importance of self regulated learning and structured time management (Zimmerman, 2002), there is a noticeable gap between theoretical recommendations and practical technological solutions tailored to students' daily academic planning needs.

The proposed project addresses this gap by developing a web based system that integrates rule based Artificial Intelligence logic specifically for academic task prioritization and adaptive scheduling. Unlike conventional task managers, the system computes priority scores based on measurable criteria such as deadline proximity, task complexity, and workload distribution. Unlike LMS platforms, it is designed explicitly as a decision support tool rather than a content management system. Unlike AI tutoring systems, it focuses on operational academic organization rather than instructional personalization.

The unique contribution of this project lies in its integration of transparent, rule-based prioritization with dynamic schedule generation tailored to university students. By combining principles of decision support systems (Power, 2002), self-regulated learning theory (Zimmerman, 2002), and contemporary AI enabled system design, the project provides a context-aware productivity tool that bridges the gap between academic theory and practical student needs.

In doing so, the system offers both localized relevance within Kenyan higher education and broader applicability to institutions seeking structured, student-centered academic productivity solutions.

CHAPTER THREE

METHODOLOGY

This chapter describes the methodological approach used to design and develop the proposed AI-powered student productivity system. It explains how the solution was conceptualized, the research strategy adopted, the resources utilized, and the processes followed in system development.

The project adopts a Design Science Research (DSR) methodology, which is appropriate for Information Systems projects aimed at developing innovative technological artifacts to solve identified real world problems (Hevner et al., 2004). Design Science focuses on the creation and evaluation of artifacts that address practical challenges. In this context, the artifact developed is a web based academic productivity system integrating rule based Artificial Intelligence logic.

The solution was conceptualized through three primary foundations:

First, *findings from the literature review* revealed that existing productivity systems either rely on manual prioritization or focus on instructional delivery rather than intelligent academic scheduling. Although AI has been widely implemented in tutoring systems and learning analytics, limited systems directly address structured academic workload management.

Second, *observation of student academic behavior*, including personal academic experience, identified recurring challenges such as overlapping deadlines, ineffective prioritization, procrastination, and increased stress during peak academic periods.

Third, *analysis of existing systems* demonstrated a gap between theoretical recommendations on self-regulated learning and practical technological tools that actively support daily academic planning.

Based on these findings, a rule based intelligent scheduling system was identified as the most suitable and feasible solution within the project scope.

PRELIMINARY INVESTIGATION

The preliminary investigation phase focuses on identifying user needs and translating them into system requirements. This ensures that the proposed system responds to real academic challenges rather than assumptions.

REQUIREMENTS ELICITATION METHODS

To gather reliable and valid user requirements, multiple elicitation techniques will be used:

a) Questionnaires

Structured questionnaires will be distributed to university students to collect quantitative data regarding:

- Current task management methods
- Frequency of missed or delayed assignments
- Challenges in prioritizing academic tasks
- Preferred features in a productivity system

This approach provides measurable insights into common user needs.

b) Semi-Structured Interviews

Interviews will be conducted with selected students to gain deeper qualitative insights into:

- How students determine task priority
- Difficulties experienced during examination periods
- Limitations of existing digital tools

Interviews allow exploration of user experiences beyond structured survey responses.

c) Observation

Observation of student planning habits will help identify behavioral patterns such as procrastination triggers and ineffective time allocation strategies.

The use of multiple methods ensures triangulation, thereby improving the reliability and validity of identified requirements.

Functional Requirements

Functional requirements define what the system must do.

The system shall:

1. Allow users to register and authenticate securely.
2. Allow users to create, update, and delete academic tasks.
3. Allow users to assign deadlines and difficulty levels to tasks.
4. Automatically calculate priority scores using rule-based logic.

5. Generate personalized study schedules based on computed priorities.
6. Dynamically update schedules when tasks are modified or completed.
7. Display tasks and schedules through a user dashboard.
8. Store and retrieve user data securely from the database.

Non-Functional Requirements

Non-functional requirements define how the system performs.

Usability:
The system shall provide an intuitive and user-friendly interface requiring minimal training.

Performance:
Schedule generation shall occur within an acceptable response time (e.g., under two seconds).

Security:
User credentials shall be encrypted and passwords securely hashed.

Reliability:
The system shall ensure data integrity and consistent performance without unexpected failures.

Scalability:
The architecture shall support future expansion to accommodate additional users and features.

Maintainability:
The system shall follow modular design principles to allow for easy updates and improvements.

SYSTEM DESIGN PHASE

The design phase translates identified requirements into structured technical blueprints that guide implementation.

System Architecture

The system will follow a *three-tier architecture,* consisting of:

1. ***Presentation Layer***
   Developed using React.js, responsible for user interaction and interface rendering.

2. ***Application Layer***
   Developed using Django (Python), responsible for business logic, authentication management, API communication, and AI prioritization processes.
3. ***Data Layer***
   Implemented using PostgreSQL to manage structured storage of users, tasks, and schedules.

This layered structure ensures separation of concerns, scalability, and maintainability.

AI Logic Design

The proposed system uses a rule-based prioritization algorithm to organize and schedule academic tasks. This approach was selected because academic scheduling involves structured information such as deadlines, task difficulty, and available study time. These factors can be clearly defined and evaluated using logical rules without requiring complex or probabilistic AI models.

Each task entered into the system contains specific attributes, including the deadline date and an assigned difficulty level. The system analyzes these attributes to determine how urgent and demanding each task is. Deadline proximity is used to measure urgency, meaning that tasks with closer deadlines receive higher priority. Task difficulty represents the level of effort required to complete the task. In addition, the system considers the overall workload within a given period to avoid overloading a single day or study session.

To calculate task priority, the system applies a weighted scoring formula:

Priority Score = (Urgency Weight × Deadline Proximity) + (Difficulty Weight × Task Complexity)

Tasks with higher priority scores are scheduled earlier within available study time blocks. This ensures that urgent and difficult tasks are addressed before less critical ones.

The rule-based approach offers several advantages. First, it produces consistent and predictable results, meaning the same input will always generate the same schedule. Second, it is computationally efficient and does not require advanced hardware or external AI services. Third, logic is transparent, allowing users to understand how their tasks are being prioritized.

Since academic scheduling is a structured problem with clearly defined inputs, a rule-based algorithm is sufficient and appropriate for this project. It provides reliable decision support while maintaining simplicity and clarity in system design.

DEVELOPMENT RESOURCES AND PROCESS

Technical Resources

The system will be developed using a modern full-stack architecture.

Frontend development will be implemented using *React.js* to provide a responsive and interactive user interface. Backend development will utilize the *Django framework (Python)* due to its robustness, security features, and seamless integration with rule-based logic modules. The database layer will be managed using *PostgreSQL,* selected for reliability and structured data management capabilities.

The AI prioritization engine will be implemented in *Python*, enabling structured rule-based computation. Development will be conducted using *Visual Studio Code*, with *Git and GitHub* used for version control and systematic code management. API testing and integration validation will be performed using *Postman* and browser-based developer tools.

These technologies were selected based on scalability, compatibility, maintainability, and suitability for integrating structured AI logic within a secure web-based system.

System Development Process

The system will be developed through iterative stages:

1. Requirements analysis
2. System design (DFD, ERD, architectural modeling)
3. Backend development
4. Frontend development
5. AI logic implementation
6. Full system integration
7. Testing and debugging
8. Deployment and documentation

An Agile-inspired iterative approach will be adopted to allow incremental testing, continuous evaluation, and progressive refinement. This ensures early identification of issues and alignment with user needs throughout development.

CHAPTER FOUR

SYSTEM ANALYSIS AND DESIGN

This chapter presents the analysis and structural design of the proposed AI-powered Student Productivity System. The purpose of this chapter is to translate the system requirements identified in Chapter Three into logical and technical representations that will guide development. The chapter outlines the system's data flow structure, database design, and implementation timeline. These design models ensure that the proposed solution is logically consistent, scalable, and aligned with user requirements.

System analysis focuses on understanding how information moves within the system and how different components interact to achieve desired functionality. System design transforms these analytical findings into structured diagrams that serve as blueprints for implementation.

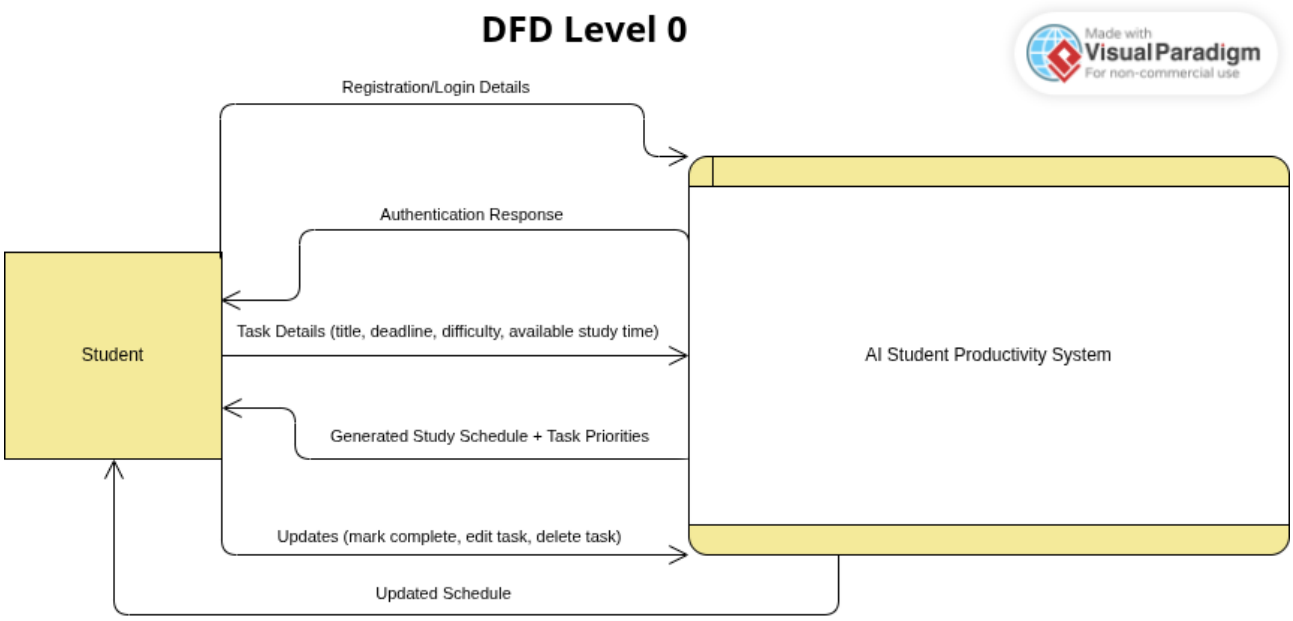
DATA FLOW DIAGRAM (DFD)

The Data Flow Diagram illustrates how data moves through the system, including user inputs, internal processing activities, data storage, and system outputs. It provides a structured representation of how the proposed application will operate.

Context Diagram (Level 0 DFD)

The Context Diagram represents the entire system as a single process interacting with an external entity. In this system, the external entity is the Student, who interacts with the AI Student Productivity System. The student provides registration details, login credentials, and academic task information such as task title, deadline, and difficulty level. The system processes this information internally using rule-based prioritization logic and generates a personalized study schedule. The generated schedule and computed task priorities are then presented to the student through the user's dashboard.

The Context Diagram demonstrates that the system functions as an intelligent intermediary between user-provided academic data and structured scheduling outputs.



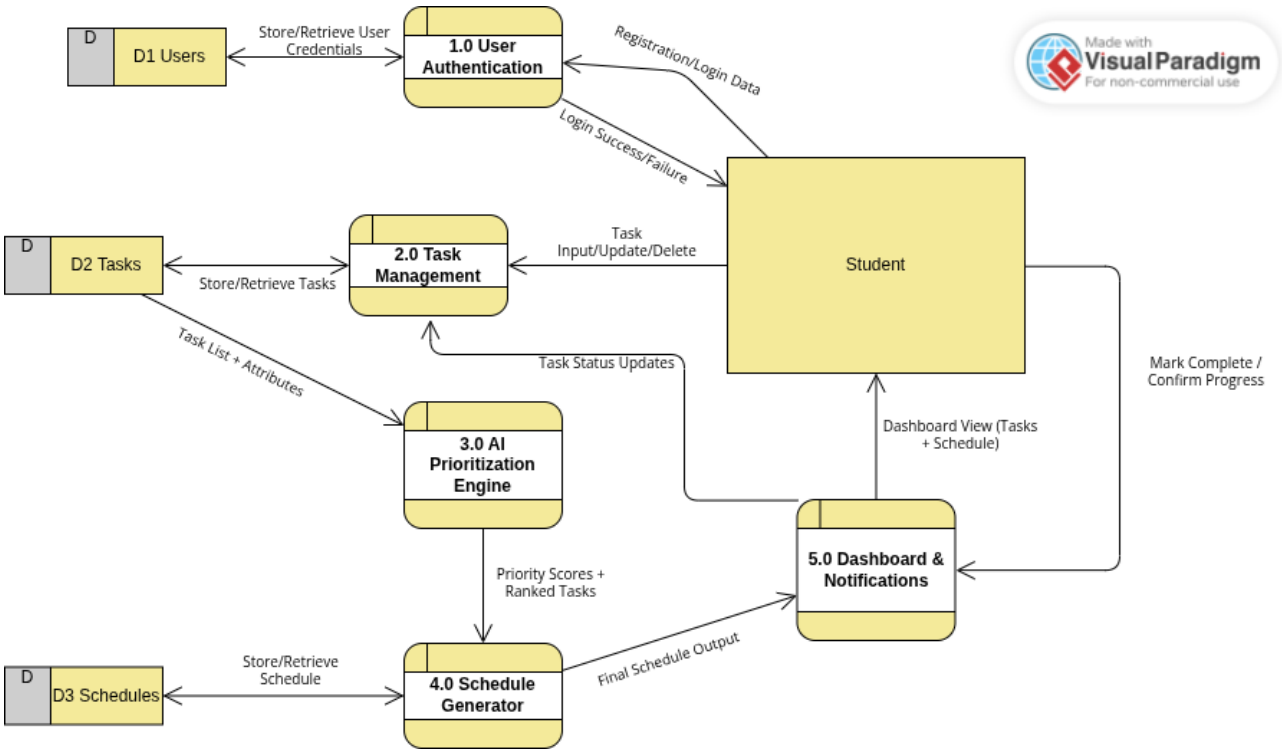
Level 1 Data Flow Diagram

The Level 1 DFD expands the system into major internal processes to show how data is handled in greater detail. The system is divided into five primary processes: User Authentication, Task Management, AI Prioritization Engine, Schedule Generator, and Dashboard Display.

The User Authentication process verifies login and registration data against the User Database. Once authenticated, the student accesses the Task Management process, where tasks can be created, modified, or deleted. Task information is stored in the Task Database.

The AI Prioritization Engine retrieves task data and computes priority scores based on deadline proximity and difficulty level. The ranked tasks are then forwarded to the Schedule Generator, which assigns study sessions within available time blocks. The resulting schedule is stored in the Schedule Database and displayed to the student through the Dashboard interface.

This structured decomposition ensures clear separation of system responsibilities and enhances maintainability.



ENTITY RELATIONSHIP DIAGRAM (ERD)

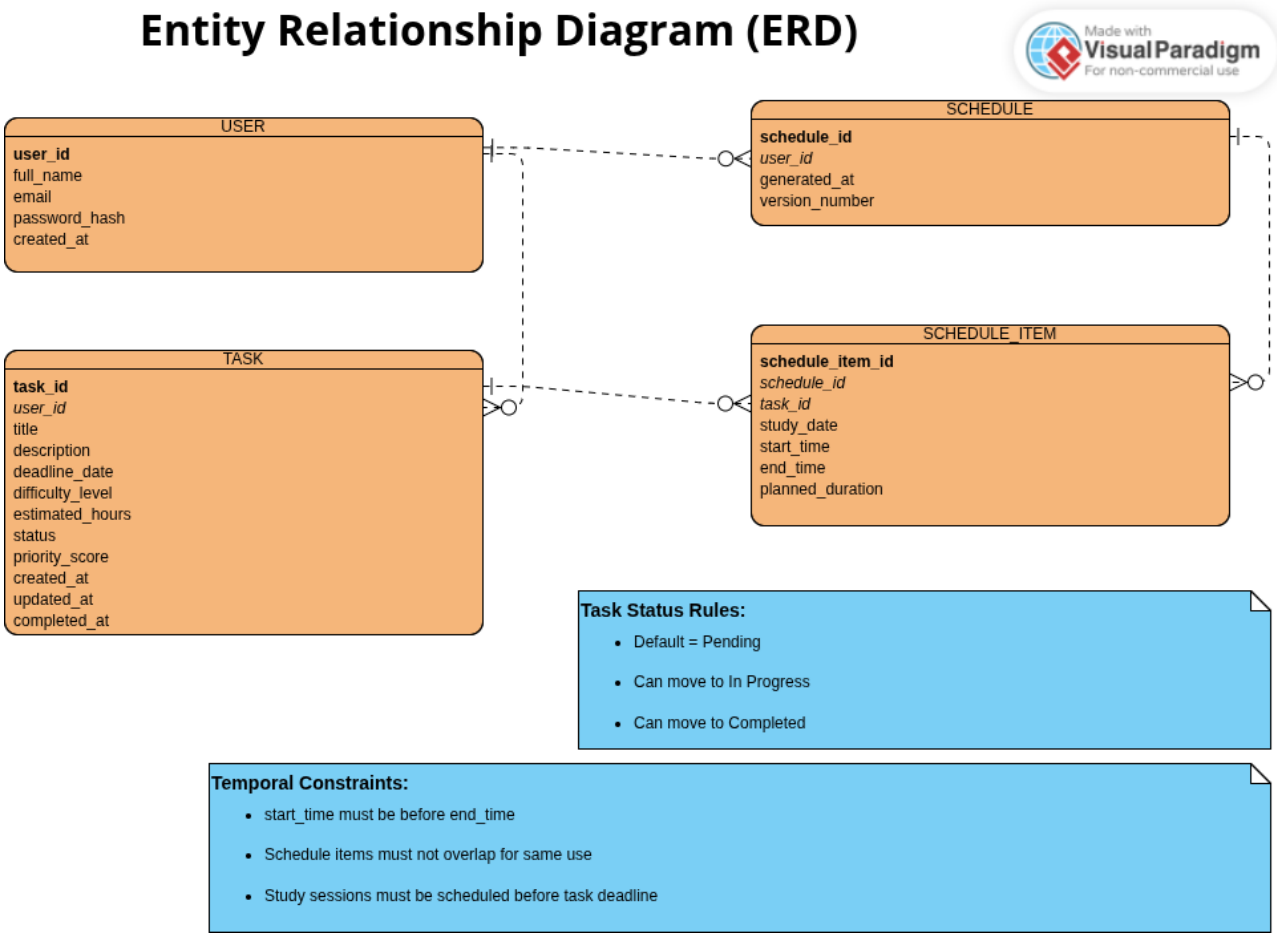
The Entity Relationship Diagram defines the database structure of the system and illustrates how entities relate to one another. The ERD ensures data integrity, consistency, and normalization within the application.

The system consists of four primary entities: User, Task, Schedule, and Schedule_Item.

The User entity contains attributes including user_id as the primary key, full_name, email, password_hash, and created_at. The Task entity includes task_id as the primary key, user_id as a foreign key referencing the User entity, title, description, deadline_date, difficulty_level, estimated_hours, status, priority_score, created_at, updated_at, and completed_at. The Schedule entity contains schedule_id as the primary key, user_id as a foreign key, generated_at, and version_number. The Schedule_Item entity includes schedule_item_id as the primary key, schedule_id as a foreign key referencing Schedule, task_id as a foreign key referencing Task, study_date, start_time, end_time, and planned_duration.

The relationships among these entities are structured as follows. One User can create many Tasks, establishing a one-to-many relationship. One User can generate multiple Schedules over time. Each Schedule contains multiple Schedule_Items, allowing several study sessions within one schedule. A

single Task may appear in multiple Schedule_Items, enabling tasks to be divided into several study blocks when necessary.



System Constraints and Logical Conditions

To ensure data integrity and logical consistency, several operational constraints are defined within the system design. Tasks are created with a default status of Pending and may transition to In Progress and subsequently to Completed. Once a task is marked as Completed, it is not expected to revert to a previous status unless explicitly modified by the user.

Study sessions generated by the system must be scheduled before the corresponding task deadline. In addition, scheduled time blocks for a single user must not overlap, and the start_time of any study session must be earlier than its end_time. The priority_score attribute is stored within the Task entity and recalculated whenever task attributes such as deadline or difficulty level are updated, or when a new schedule is generated.

These structural conditions ensure that the system maintains consistency, prevents scheduling conflicts, and supports reliable task prioritization.

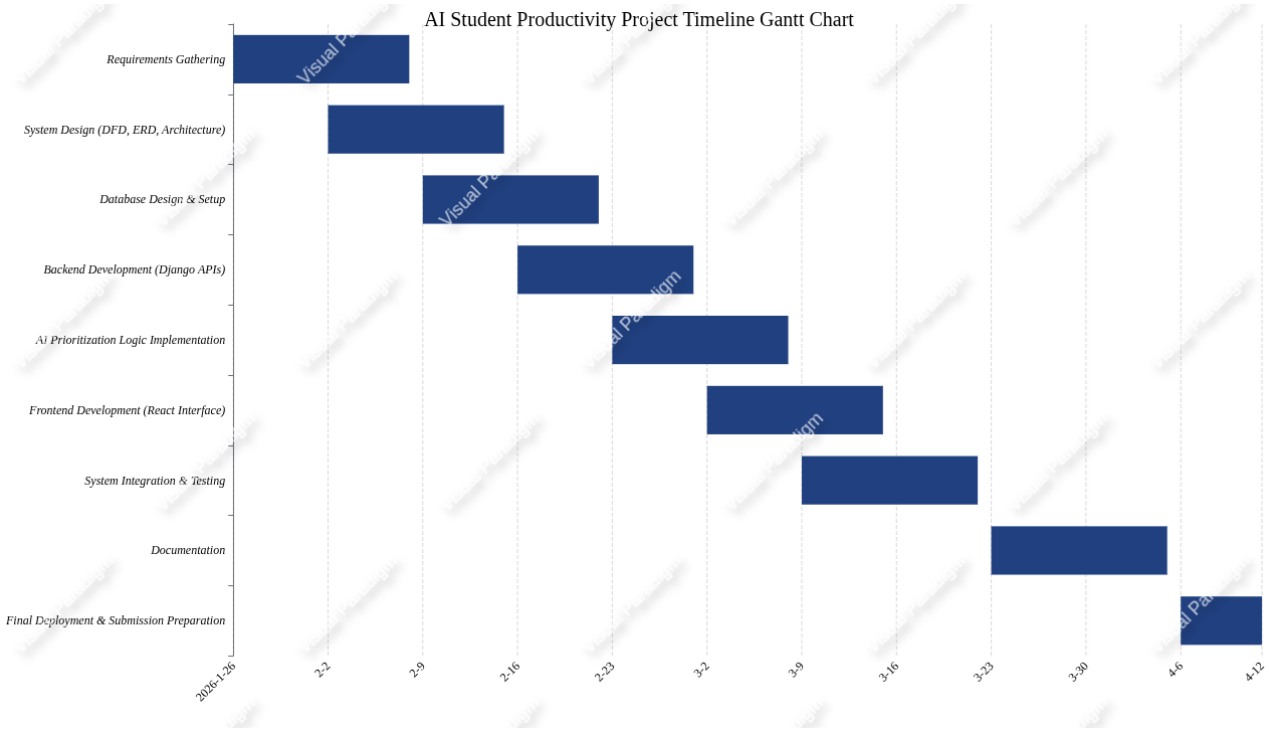
PROPOSED TIMELINE

The implementation of the proposed system is scheduled to run from 26 January 2026 to 12 April 2026. The project timeline is structured to allow sequential yet overlapping development phases to ensure efficiency and iterative improvement.

The development process begins with requirements gathering and user analysis, followed by system design activities including the creation of Data Flow Diagrams and the Entity Relationship Diagram. Database setup and backend development are implemented subsequently to establish core functionality. The AI prioritization logic is integrated after the task management module is operational. Frontend development is conducted alongside backend stabilization to enable early integration testing.

System integration and testing are scheduled in the later phase of the timeline to validate functionality, performance, and data consistency. Documentation and final deployment preparation conclude the project to ensure readiness for submission.

The proposed timeline reflects an iterative development approach that allows for continuous refinement and progressive system validation throughout the semester.



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